

The Current Status of Computing

- **Context:** With the rapid increase of mobile devices and the ability to connect things, humans are in a post-PC era where connected devices are replacing traditional desktop and laptop computers.

Modern Computing Technologies

Cloud Computing

- **Definition:** Application of hardware and software to deliver readily available, on-demand, and measured computing services over the Internet, making IT services as convenient as household utilities.
- **Mechanism:** Data is stored in a "cloud" (symbolizing the Internet) rather than physical local devices, allowing users to open files and applications anywhere via any internet-connected device.
- **Applications:**
 - *Cloud Storage:* Google Drive and iCloud for backing up and restoring local files.
 - *Translation:* Google Translate for instant translation of words, dialects, phrases, and web pages across over 100 languages.
 - *Streaming Services:* Netflix, Apple TV, and Deezer for managing real-time data logs.

Virtualization

- **Definition:** Technology that simulates hardware functions by creating multiple virtual machines on a physical server acting as a hardware device. Detaches software from hardware by converting physical devices into logical folders or files.
- **Applications:**
 - *Server Virtualization:* Companies use one server to support different functions for structured resource allocation.
 - *Desktop Virtualization:* Users run several desktop operating systems from a single computer.
 - *Storage Virtualization:* Allows connected devices, servers, and applications to access information from a centralized dashboard like a cloud.
- **Relationship to Cloud:** Cloud computing uses virtualization technology to achieve its purpose.

Internet of Things (IoT)

- **Definition:** A network of an extensive range of physical objects that automatically share data over the Internet, enabling faster information exchange and communication between "things" (e.g., GPS, laser scanners) to accept, locate, monitor, and manage them.
- **Key Technologies:**
 - *RFID (Radio Frequency Identification):* Electronic tags and smart cards.

- *Sensor Technology*: Detects and collects environmental signals (temperature, humidity, speed).
- *Embedded Systems*: Integrated hardware and software providing specific functions (ATMs, digital cameras, calculators).

Big Data

- **Definition**: Massive and complex datasets characterized by high volume, variety of data types, and velocity as data is collected and analyzed at ever-increasing rates.
- **Evolution**: Shifted from small early IT volume (books/newspapers in physical libraries) to digitization (e-books, online news).
- **Applications**:
 - *Banking*: Tracking financial market activities and using network analytics and natural language processing to catch illegal trading.
 - *Healthcare*: Acquiring data from thousands of patients to enable evidence-based medicine.
 - *Education*: Measuring teacher effectiveness to ensure pleasant student and teacher experiences.

Artificial Intelligence (AI)

- **Definition**: Enables machines to learn from experience, adjust to new inputs, and perform human-like activities. Regarded as technology that can change the world or end it.
- **Synergy & Ecosystem**: Combines with cloud computing and big data to drive digital transformation. Cloud computing is the foundation of AI; IoT provides the raw materials used to produce the fertilizer that Big Data feeds into AI.
- **Applications**: Facial recognition (unlocking smartphones), voice recognition, chatbots, robots, virtual assistants (Siri, Alexa), and e-payments.

Other Modern Computing Technologies

- **Fog Computing**: Idea proposed by Cisco expanding on cloud computing. Stores data and applications in edge devices close to the ground (near end-users) rather than remote clouds in the sky, allowing local data analytics and real-time data distribution for factories, streetlights, and commodities.
- **Edge Computing**: Decentralized, distributed cloud infrastructure close to data sources where raw data is analyzed locally. Predicted as the natural progression from cloud to edge and the next big thing, playing a critical role in advancing IoT.

Impacts of Computing Across Industries

- **Overview**: Powerful, inexpensive, easy-to-use computers affect entertainment, productivity, relationships, work, and lifestyles.
- **Business Organizations**: Uses tools like Microsoft Excel and SQL Databases to organize data, reduce human error, minimize physical file cabinets, and simplify error tracking for better decisions.

- **Banking Industries:** Improves transactions via ATMs (allowing deposits/withdrawals anytime/anywhere) and allows bank tellers to quickly look up and manage customer accounts by name.
- **Education:** Delivers accessible, high-quality education through innovative approaches like blended learning, online degrees, and courses.
- **Healthcare:** Stores patient treatment records and medical history in databases for doctor access and utilizes medical software/hardware for faster, easier diagnosis.
- **Government:** Streamlines public information storage to process paper and service requests faster, more reliably, and with greater accuracy.
- **Robotics:** Integrates computing with science and engineering. Industrial robots handle complex tasks, while physical humanoid robots like Sophia (created by Hanson Robotics, activated Feb 2016) demonstrate human-like communication and expression capabilities.
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